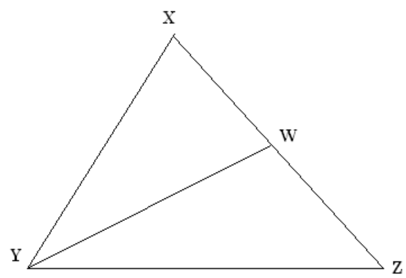
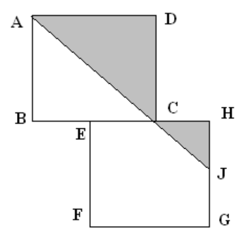
JMO – 2011

1. A Shopkeeper enhances the cost price of an article by 50% and while selling allows two successive discounts of 20% and 30% to the customer. Determine the profit or loss on the article.
2. There are two containers out of which one contains exactly 16 litres of water and the other container contains exactly 9 litres of water. There is no measuring mark on the containers. Using only the two containers determine how to bring 6 litres of water from a pond?
3. In a bag there are 13 red, 5 blue, 1 white anf 9 green balls. Without looking in to the bag determine the least number of balls that can be drawn such that three of them are of the same color.
4. Find how many positive integers n with $n \le 100$, $n^3 + 5n^2$ is a square of an integer.
5. In the diagram ΔXYZ is isosceles with XY = XZ. Also, point W is on XZ so that $XW = WY = YZ$. Determine the measure of $\angle XYZ, \angle XYW$.



6. Determine how many positive integers less than 400 can be created using only the digits 1, 2 or 3 with repetition of digits allowed.
7. Squares ABCD and EFGH are equal in area. Vertices B, E, C and H on the same line. Diagonal AC is extended to join, the midpoint of GH. Determine ratio of the shaded portion of the two squares.



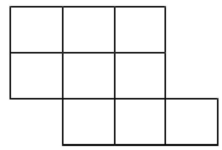
8. A and B start from two places X and Y respectively that are 60 KM apart. If A and B travel at a uniform speed of 80 MK and 20 Km per hour respectively and return back immediately to their respective places after reaching Y and X respectively, then how many often and at what distance from X, shall they meet?
9. Two numbers a and b with $0 \le a \le 1$, $0 \le b \le 1$are chosen at random. The number c is defined by $c = 2a + 2b$. The numbers a, b and c are each rounded to the nearest integers A, B and C respectively. What is the probability that $C = 2A + 2B$?
10. I bought a new plant for my garden, Anika said it was red rose, Bill said it was a purple daisy and Cathy said it was a red dahlia. Each person was correct in stating either the color or the type of plant. What was the plant that I bought?
11. A palindrome is a positive integer that is same when we read forward or backward. Find the difference between the smallest 3 digit palindrome and greatest 3 digit palindrome.
12. For which positive integral value of m, $\frac{1}{2m^2 + 15}$ will be closest to 0.01?

13. In the addition of three digit numbers shown, the letters A, B, C, D, E each represent a single digit.

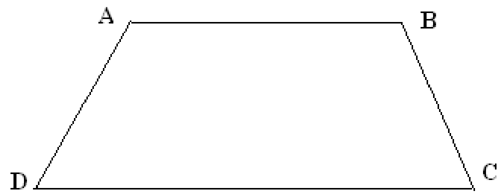
$$\begin{array}{r} A\ B\ E \\ A\ C\ E \\ +\ A\ D\ E \\ \hline 2\ 0\ 1\ 1 \end{array}$$

Find the value of $A+B+C+D+E$.

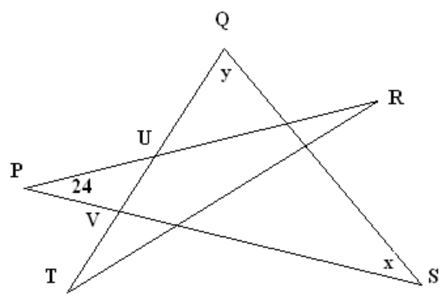
14. From the figure shown, three of the nine squares are to be selected. Each of three selected squares must share a side at least one of the other two selected squares. Find in how many ways this can be done?



15. In the trapezium ABCD, AC = BD. Show that AD = BC.



16. Let n be the largest integer for which 14n has exactly 100 digits. Counting from roght to left, find the 68^{th} digit of n.
17. In the diagram, PR, PS, QS, QT and RT are straight line segments. QT intersects PR and PS at U and V respectively. If PU = PV, $\angle UPV = 24^0, \angle PSQ = x^0, \angle TQS = y^0$, find the value of $x+y$.



18. John lists the integers from 1 to 20 in increasing order. He then erases the first half of the integers in the list and rewrites them in order at the end of the second half of the list. Find the integer in the new list which has exactly 12 integers to its left.
19. A swimmer is swimming up a river against the current. Next to bridge A he loses an empty water bottle. Twenty minutes later he notices the loss and returns around. He finds the water bottle by bridge B. Knowing that the distance between two bridges is 2 KM. How fast is the river flowing?
20. Begin with any two digit number and multiply the two digits together. If the result product a two digit number, then repeat the process. When this process is repeated all two digit numbers will eventually become a single digit. Once a product results in a single digit, the process stops.
    (a) Beginning with the number 68, determine the number of steps required for the process to stop.
    (b) Determine all two digit numbers for which the process stops at 8 after 2 steps.
    (c) Determine all two digit numbers for which the process stops at 4.
    (d) Determine a two digit number for which the process stops after 4 steps.

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